

Soru 1:

a) Valans elektron

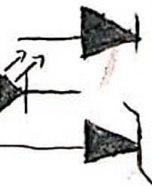
b) Saf silisyum maddesine atomların en dış yörüngesinde 5 elektron bulunan arsenik maddesi katılırsa, silisyum atomları, komşu arsenik atomlarının en dış yörüngesindeki 5 elektronun 4 tanesiyi kovalent bağ kurmasıyla p tipi malzeme; silisyum maddesine, bor maddesi enjekte edildiğinde, bor atomlarının en dış yörüngelerindeki 3 elektron, silisyum atomlarıyla kovalent bağ kurmasıyla p tipi malzeme oluşur.

c) Elektriksel iletkenlik bakımından yalıtkenlerle iletkenler arasında kalan maddelerdir. İletkenler tamamıyla iletken, yalıtkenler tamamıyla yalıtken olduğu için devrelerde kontrol elemanları olarak kullanılabilirler.

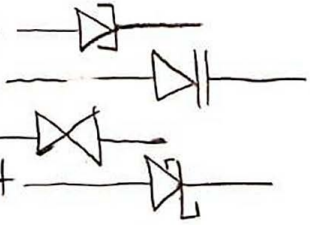
d) Diyot, Zener diyot, Tünel diyot, Varikap diyot, Şotki diyot, Led diyot, Infraruj Led diyot, Foto diyot, Optokuplörler, Transistörler, Triyak, Diyak, Tristör

e) Ters gerilimde belli değerlerde diyot ters yönde de akım akıtabildiği değerler. Ters akıma bölgesidir.

f) Kristal diyotlar
Led diyotlar
Zener diyotlar



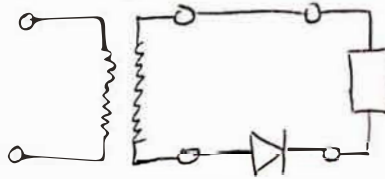
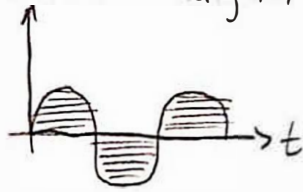
Tünel diyot
Varikap diyot
Gunn diyot
Schottky diyot



g) Alternatif akımı doğru akıma çevirir.

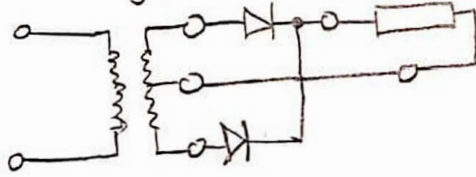
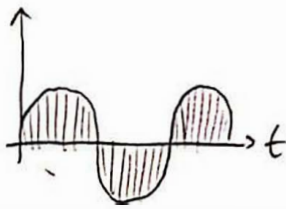
Yarım Dalga Doğrultma

Alternatif dalganın pozitif veya negatif tarafı engellenir.



Tom Dalga Doğrultucu

Doğru akım dalgalarını kayıpsız olarak geçirir.



h) Girişine uygulanan sinyalin bir bölümünü kırpan devredir.
Devrede istediğimiz dalga şeklini almak için kullanılır.

i) Alternatif sinyali farklı bir doğru akım seviyesine çıkartıp
kullanılan devredir.
Çıkışta kayma istendiğinde kullanılır.

j) Ters polarizasyonda yapıldığı takdirde devreye uygulanan
gerilim zener gerilimine gelene kadar güç devresini çalıştırmaz ve
devreye uygulanan gerilim zener gerilimini sabit tutar.

k) üst kolları

10^2	dekaohm
10^3	hectoohm
10^3	kiloohm
10^6	megaohm
10^9	gigaohm
10^{12}	teraohm
10^{15}	petaohm
10^{18}	exaohm
10^{21}	zettaohm
10^{24}	yottaohm

Ast kolları

10^{-1}	desiohm
10^{-2}	santiohm
10^{-3}	milliohm
10^{-6}	microhm
10^{-9}	nanoohm
10^{-12}	picoohm
10^{-15}	femtoohm
10^{-18}	attoohm
10^{-21}	zeptoohm
10^{-24}	yoktoohm

l) Güney, istenmeyen dalgaları, sızımları, gürültüleri engellemektir.

Sembolü: $\text{---} \perp \text{---}$

Birimi : Frad

10^{-3}	: milifarad
10^{-6}	: microfarad
10^{-9}	: nanofarad
10^{-12}	: picofarad

m) En az iki elektrik devresini elektromanyetik indüksiyonla birbirine bağlar. Elektrik devreleri arasında elektromanyetik alan ile enerji nakleder. Elektrik enerjisinin belirli güçünde gerilim ve akımlarda değişim yaparlar.

$$2) \quad I = \frac{14,3 \text{ V}}{R_T} = \frac{14,3 \text{ V}}{6,8 \text{ k}\Omega} = 2,1 \text{ mA}$$

$$V_{R_1} = R_1 \cdot I = 4,6 \cdot 2,1 = 9,66 \text{ V}$$

$$V_{R_2} = R_2 \cdot I = 2,2 \cdot 2,1 = 4,62 \text{ V}$$

$$V_o = V_{R_2} - 5 \text{ V} = 4,62 - 5 = -0,38 \text{ V}$$

$$3) \quad V_{e_f} = V_{D_2} = V_1 = 0,7 \text{ V}$$

$$I_1 \cdot 2,2 = 0,7$$

$$\boxed{I_1 = 0,31 \text{ mA}}$$

$$E = V_{e_f} + V_{D_2} + I_2 \cdot R_2$$

$$25 = 0,7 + 0,7 + I_2 \cdot 4,6$$

$$\boxed{I_2 = 5,13 \text{ mA}} = I_{D_1}$$

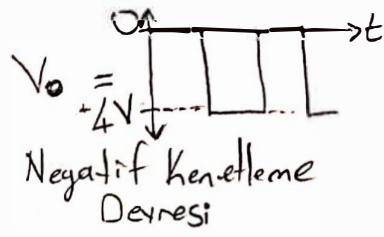
$$\boxed{I_{D_1} = 5,13 \text{ mA}}$$

$$I_{D_2} = I_2 - I_1$$

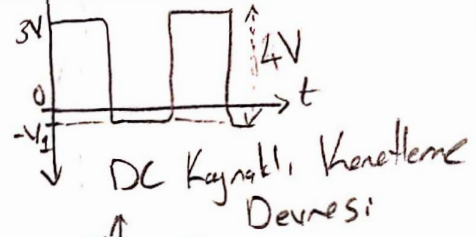
$$I_{D_2} = 5,13 - 0,31$$

$$\boxed{I_{D_2} = 4,82 \text{ mA}}$$

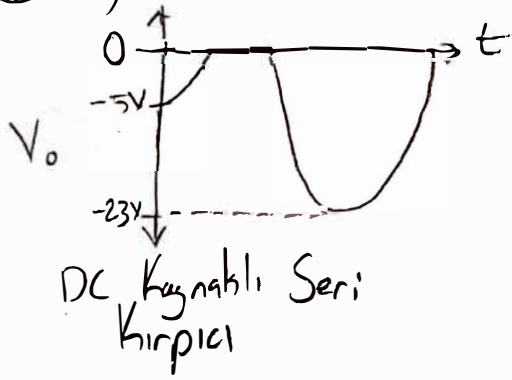
⑤ a)



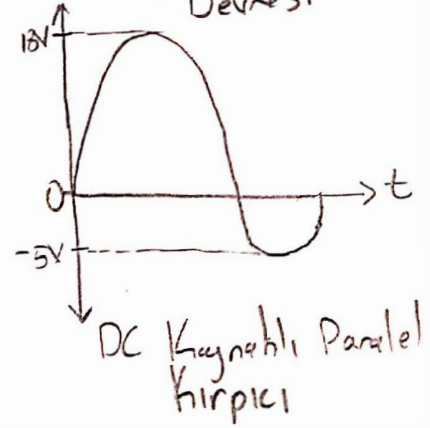
b)



④ a)



b)



$$\textcircled{6} \quad a) \quad V_{i, \min} = \frac{V_z (R_L + R_s)}{R_s} = \frac{6,8 (2,2 \cdot 10^3 + 220)}{2,2 \cdot 10^3} = \boxed{7,48 \text{ V}}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_z}{R_L} = \frac{6,8}{2200} = 3,09 \text{ mA}$$

$$I_{RS} = I_{z_{\max}} + I_L = 10 + 3,09 = 13,09 \text{ mA}$$

$$V_{i, \max} = (13,09 \cdot 10^{-3}) (2,2 \cdot 10^3) + 6,8 = \boxed{9,68 \text{ V}}$$

7,48 V ve 9,68 V arasında çalışır.

$$b) \quad R_L = \frac{V_z \cdot R_s}{V_i - V_z} = \frac{12 \cdot 10}{50 - 12} = \boxed{3,15 \Omega} \text{ min}$$

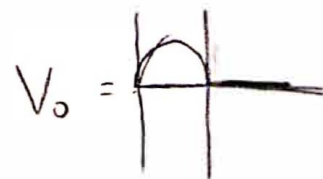
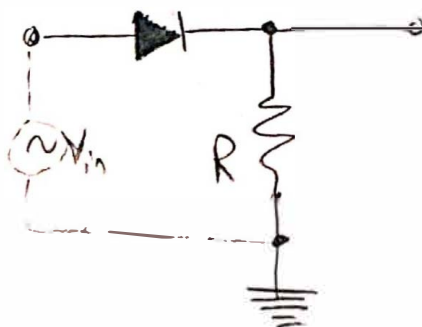
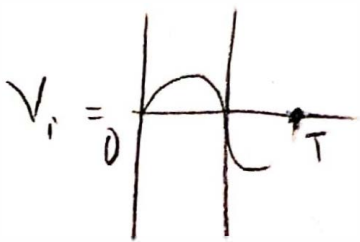
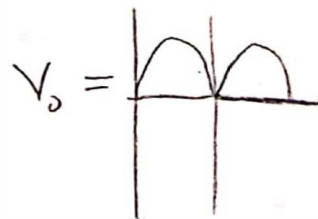
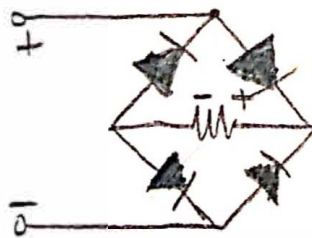
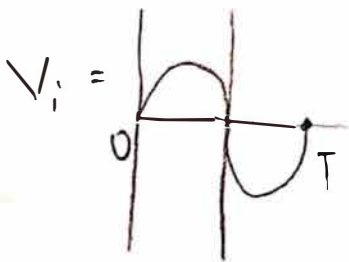
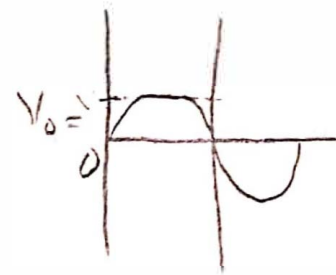
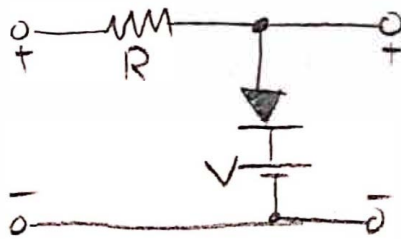
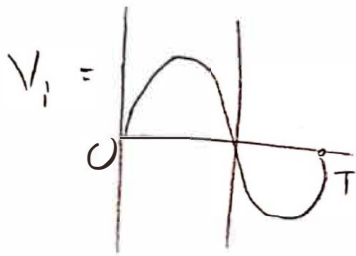
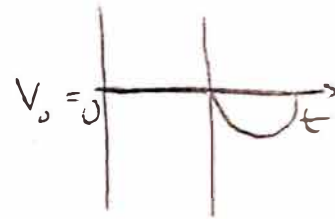
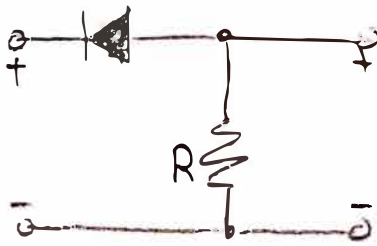
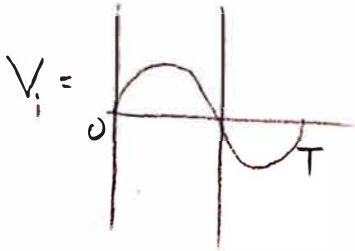
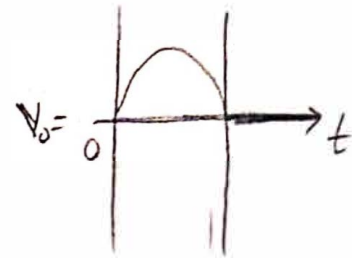
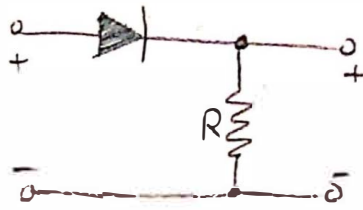
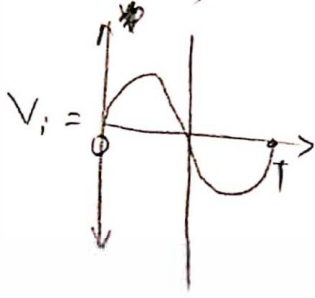
$$V_{RS} = 50 - 12 = 38 \text{ V} \quad I_{RS} = \frac{38}{10} = 3,8 \text{ A}$$

$$I_L = 3,8 - 0,06 = 3,74 \text{ A} \quad R_L = \frac{12}{3,74} = \boxed{3,21 \Omega} \text{ max}$$

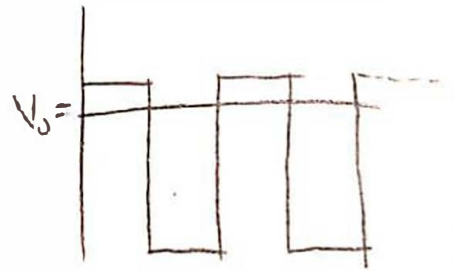
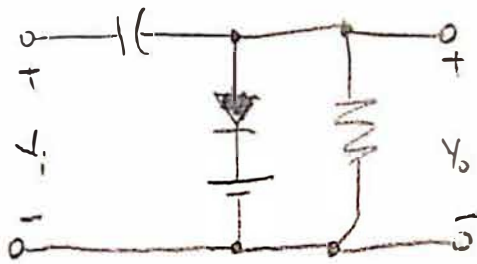
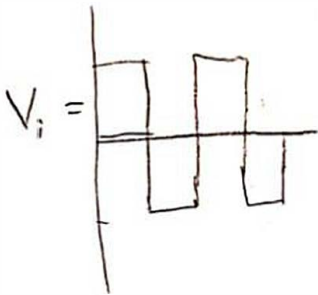
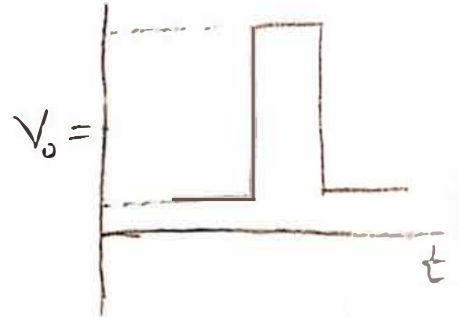
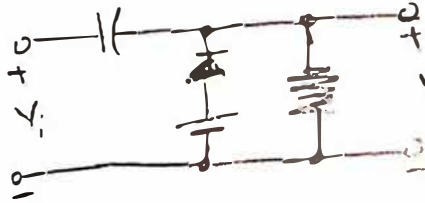
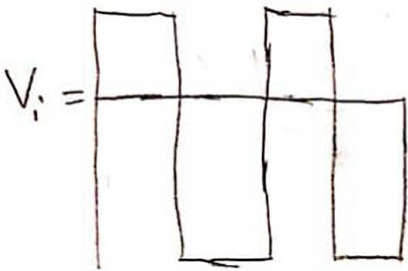
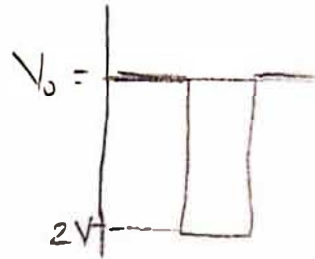
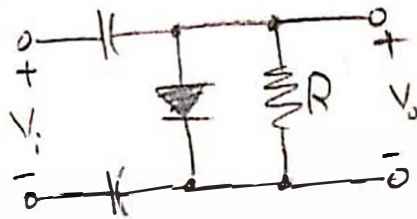
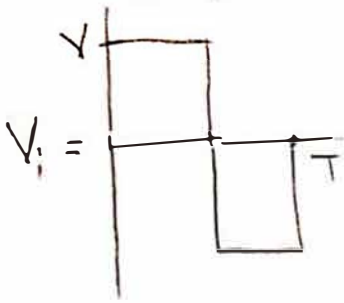
3,15 Ω ile 3,21 Ω arasında çalışır.

VİZE ÖDEV

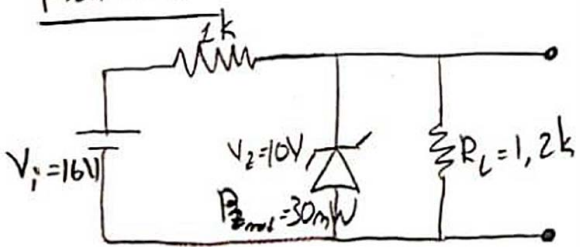
Hafıza



Haft 8



Haft 9



$$V_z = \frac{16V}{1k + 3k} \cdot 3k = 12V = V_z$$

$$V_i = V_R + V_z$$

$$V_R = V_i - V_z = 16V - 10V = 6V = V_R$$

$$I_R = I_z + I_L$$

$$\frac{V_i - V_z}{I_k} = I_z + \frac{10V}{3k}$$

$$6mA = I_z + 3.33mA$$

$$P_z = I_z \cdot V_z = 2.67mA \cdot 10V$$

$$P_z = 26.7mW$$